

Christine Ryan

Software Engineer
(Backend / Systems)

Christine Ryan
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Professional Summary

Software Engineer with experience designing and building backend systems, APIs, and data-driven applications. Strong focus on system reliability, deterministic workflows, and clear interface boundaries. Background in Python, Java, C++, and full application lifecycle development, with experience shipping demo-ready systems and working across product and design.

Core Skills & Engineering Focus

Languages & Frameworks

Python, Java, Swift, JavaScript, SQL, HTML, CSS

Systems & Tools

Flask, HTTP APIs, MapKit, Git, SQL Server, Apache Spark, Neo4j, Tableau, Xcode, IntelliJ

Engineering Focus

Demo-ready system design and reliability
Narrative-driven technical storytelling
Graceful failure handling and fallback strategies
Audience-first UX for technical products
Real-data ingestion and deterministic workflows
Cross-functional collaboration with product and design

Demo Systems & Technical Projects

applyAI— Narrative AI Demo Platform

Python, Flask, LLM adapters, policy & boundary layers

Designed and iterated prompt-driven workflows that simulate AI behavior across capability levels while enforcing explicit constraints and consent.

Built prompt guardrails using policy layers and state machines to prevent overreach, hallucination, and unsafe interpretation.

Developed evaluation scenarios to compare prompt behaviors under controlled conditions (no-AI, shallow, bounded).

Translated ambiguous user inputs into structured prompt schemas optimized for clarity, correctness, and explainability.

Enabled graceful degradation and human-understandable failure modes for live demos.

LetsA(pp)I— Career Application Exploration Tool

Python, data ingestion, workflow orchestration

Designed prompt-based transformations that convert unstructured job data into structured, explorable narratives.

Emphasized traceability and reproducibility of AI-assisted outputs for evaluation and demonstration.

Built workflows to showcase how prompt choices affect output quality and interpretation.

OpenFlights — Data-Driven Visualization Server

C++, HTTP server, large-scale datasets, map-based UI

Used prompt-driven workflows to design, iterate, and refine backend endpoints, data ingestion logic, and UI storytelling flows.

Leveraged prompting to translate high-level visualization goals into structured system behavior and API designs.

Iteratively evaluated prompt outputs against correctness, performance, and clarity requirements before implementation.

Built a deterministic C++ backend that ingests large-scale real-world aviation datasets and serves interactive, explorable visualizations.

Designed map-based UI flows optimized for human comprehension of complex system state, informed by prompt-guided exploration.

TrainApp (StoryTracks) — Interactive iOS Demo App

Swift, MapKit, animations

Used prompt-driven exploration to design interaction metaphors (animated trains, stations, routes) that make system state immediately understandable.

Iteratively refined prompts to translate high-level UX goals into concrete UI behaviors, animations, and state transitions.

Evaluated prompt outputs for clarity, cognitive load, and child-friendly interpretation before implementation.

Implemented a deterministic iOS application that integrates real-world map data with custom visual layers informed by prompt-guided design.

Replaced verbal explanation with motion-based storytelling, reducing user confusion and increasing engagement.

Password Manager — Cross-Platform GUI Application

Java, Swing

Led development of a cross-platform GUI application prioritizing clarity, safety, and error prevention.

Designed UI flows that clearly communicate sensitive system behavior.

Contributed to overall application architecture and code organization.

Education

De Anza College — Continuing Computer Science Education

Sept 2023 - Current, Cupertino

Advanced Java, Swift iOS Development, Object-Oriented Architecture, Data Structures, Assembly Language

San José State University — B.S. Management Information Systems

Sept 2012 - May 2017, San Jose

Honors: Dean's Scholar (multiple terms)

Additional Experience

Swim Instructor / Lifeguard — Menlo Swim & Sport, YMCA

Delivered live instruction to diverse audiences in safety-critical environments.

Adapted explanations in real time based on participant understanding and conditions.

Developed composure, situational awareness, and clear communication under pressure.